OPERA Analysis Plan

Outcome-guided Patient Embedding for Registry Adaptation

Working document — June 2, 2026

Scientific Question

Core question

How should a general EHR foundation model be adapted to a specialised haematology registry to maximise outcome prediction across multiple diseases and outcomes simultaneously — and does joint multi-disease training produce a single model that outperforms disease-specific models?

The question is motivated by the DALY-CARE research programme:

1. **DALY-CARE** established the infrastructure for observing patients across haematological malignancies in a unified registry.
2. **HemaSphere paper** showed empirically that joint training across lymphoma subtypes improves prediction — diseases share structure.
3. **OPERA** asks: can we make this cross-disease sharing principled and measurable, using outcomes as supervision signals?

The key intellectual claim is that *outcome trajectories encode clinical knowledge that clinical presentation alone does not capture*, and that a foundation model adapted with this signal will generalise better than any tabular model trained on disease-specific features.

Experimental Design Overview

The experiment is a full factorial over three axes:

Axis	Levels	Script
Cohort scope	Per-cohort vs. all cohorts jointly	`finetune.py` / `joint_finetune.py`
Model variant	5 tiers (see §??)	`sweep.py`
Outcome	5 outcomes × 2–3 windows	`sweep.py`

All results flow through `opera/run/sweep.py`, which produces a single `results_table.csv` and `results_table_auroc.tex` directly usable in the paper.

Cohorts and Outcomes

Disease Cohorts

Cohort	Abbreviation	Notes
Diffuse large B-cell lymphoma	DLBCL	Primary benchmark (HemaSphere)
Chronic lymphocytic leukaemia	CLL	Slow natural history
Myelodysplastic syndrome	MDS	Competing risks prominent
Multiple myeloma	MM	Heavy treatment heterogeneity
Follicular lymphoma	FL	Long follow-up

Outcomes

Outcome	Windows	Rationale	Cohorts
All-cause mortality	1y, 2y	Primary survival endpoint; clean signal	All
Treatment failure	6m, 1y, 2y	Disease control; DLBCL primary target	DLBCL, CLL, MM
Acute kidney injury	7d, 30d	Treatment toxicity; shared mechanism	All
Severe infection	30d, 90d	Immunosuppression; cross-disease signal	All
Adverse events (composite)	30d, 90d	Safety profile	All

Note on IPI baselines

For each cohort where an IPI-family score is available (NCCN-IPI for DLBCL, FLIPI-2 for FL, CLL-IPI for CLL), compute AUROC from the pre-computed score column in the population CSV. This is a *competing baseline*, not a training input. Report as a separate column in every results table.

Model Variants

The experimental design has five tiers, designed so that each tier answers a specific question by comparison with the tier below.

Tier structure and contrasts of interest

Tier 0 — Clinical baseline (IPI) NCCN-IPI / CLL-IPI / FLIPI-2 computed from RKKP variables. *Contrast*: Does any ML model beat the clinical standard?

Tier 1 — Tabular ceiling (with RKKP) XGBoost on 87 features including RKKP quality registry variables. Trained per cohort per outcome. *Contrast*: What is the ceiling when all structured data is available?

Tier 2 — Tabular floor (EHR only) XGBoost on EHR-derived features only, no RKKP. Trained per cohort per outcome. *Contrast*: What does a foundation model need to beat to justify its complexity?

Tier 3 – Foundation model pretraining ladder Shared encoder, evaluated under explicit pretraining conditions:

(a) **No pretraining / finetune only** – randomly initialised BONSAI architecture finetuned within each cohort. This is a sanity floor, not expected to be competitive.

(b) **General pretrain only** – general-population EHR encoder, no haematology-specific adaptation.

(c) **Haematology-only pretrain** – MLM pretraining only on the DALY-CARE haematology cohort. This tests whether the extra general population patients add clinically useful structure beyond scale within haematology.

(d) **DAPT** — continued MLM pretraining on DALY-CARE. Tests domain shift correction alone.

Tier 4 – OPERA outcome-guided adaptation Survival-informed contrastive adaptation, evaluated under two scopes:

(a) **OPERA (contrastive, per-cohort)** — DAPT + survival-informed multi-outcome contrastive adaptation. Tests outcome-guided representation learning within a single disease.

(b) **OPERA (contrastive, joint)** — same model trained on all cohorts simultaneously. Tests cross-disease transfer.

Tier 5 – Joint multi-task finetuning Single model finetuned on all cohorts and all outcomes jointly. Evaluated per cohort $\times$ outcome cell. *Contrast*: Does joint finetuning further improve on separately finetuned OPERA? This is where the foundation model's structural advantage over tabular models is most direct.

Why each contrast matters

Comparison	What it answers	Expected direction
General pretrain > no pretraining	Does the foundation model prior matter beyond architecture?	+
General pretrain > haematology-only pretrain	Do broad-population patients add useful structure for unusual haematology patients?	+ (especially complex patients)
DAPT > Pretrain only	Is haematology domain shift real and correctable?	+
OPERA > DAPT	Do outcomes add signal beyond what sequence distributions encode?	+
OPERA joint > OPERA per-cohort	Does cross-disease contrastive training help?	+ (especially small cohorts)
Joint finetune > per-cohort finetune	Does a single joint model beat separately trained models?	+ for FM; − for tabular
FM EHR-only > Tabular EHR-only	Is the FM better	+ (our primary bet)

The Sigma Analysis

The learned σ_k parameters from the contrastive stage are a scientific finding independent of prediction performance.

Sigma interpretation

Low σ_k → outcome k strongly structures the embedding space. High σ_k → noisy / hard to learn jointly.

Testable prediction: σ_k values from the contrastive stage should correlate with σ_k values from the joint finetune stage (both using Kendall weighting). Concordance validates that the contrastive stage learned a clinically coherent representation, not just an optimisation artefact.

The sigma analysis is a publishable result regardless of whether AUROC improves. If mortality gets low σ and adverse events get high σ, that is a finding about the latent structure of haematological disease that no tabular paper can produce.

Uncertainty and Clinician Validation

Clinician validation protocol

1. Run `extract_uncertainty()` with $N = 30$ MC Dropout samples on the held-out test set.
2. Extract the top-100 most uncertain patients using `rank_by_uncertainty(results, top_k=100)`.
3. Present each case to 2–3 haematologists (blinded to model output).
4. Ask: *"Would you want additional information before making a treatment decision?"* (binary, ecologically valid).
5. Compute point-biserial correlation between model uncertainty and clinician uncertainty ratings.

If correlation is strong: model uncertainty identifies clinically ambiguous cases. This is a usability finding — the model knows when to say "I need more information."
If correlation is weak: model is uncertain for statistical reasons that do not map to clinical difficulty. Worth reporting honestly.

Tables for the Paper

Table 1 — Main results (per cohort, primary outcome)

One row per disease cohort. Primary outcome is all-cause mortality at 1 year. Columns are model variants, metric is AUROC with 95% bootstrap CI.

Script: `sweep.py --config sweep_example.yaml` produces `results_table_auroc.tex` directly.
Filter to: `outcome == "mortality_1y"`, all cohorts, all variants.

	Clinical		*Tabular*		*Foundation model*		
Cohort	IPI	—	+RKKP	EHR only	Pretrain	DAPT	OPERA
DLBCL							
CLL							
MDS							
MM							
FL							
Mean							

AUROC (95% bootstrap CI). Bold = best per row.

Table 2 — Multi-outcome results (DLBCL)

One row per outcome and time window for the primary cohort (DLBCL). All model variants. Illustrates that the advantage pattern is outcome-dependent.

> **Filter to:** `cohort == "dlbcl"`, all outcomes × windows, all variants.

Table 3 — Joint vs. per-cohort (all cohorts, all outcomes)

Directly tests the collaborator's hypothesis: does a single joint model outperform separately trained models?

Rows: cohort × outcome. Columns: (a) per-cohort OPERA finetune, (b) joint OPERA finetune, (c) per-cohort tabular RKKP, (d) joint tabular (XGBoost with disease dummies). Metric: AUROC.

> Key comparison: column (b) vs. column (a) tests the foundation model's joint-training hypothesis. Column (d) vs. column (c) tests the same for tabular — expected to be flat or negative. The *interaction* is the finding.

Table 4 — Sigma values across outcomes

The learned σ_k values from (a) the contrastive stage and (b) the joint finetune stage.

Outcome	σ_k (contrastive)	σ_k (joint finetune)
Mortality 1y		
Treatment failure 1y		
AKI 30d		
Severe infection 90d		
Adverse events 90d		

Lower = outcome strongly structures the embedding space.

Figures for the Paper

Figure 1 — Architecture overview

A three-panel diagram showing the full OPERA pipeline:

1. **Left**: General pretraining on ~10M patients (MLM), with haematology-only pretraining as the key scale/scope contrast.
2. **Centre**: DAPT on DALY-CARE haematology cohort + OPERA contrastive stage with survival-informed soft loss and DAPT-prior cohort similarity weights.
3. **Right**: Per-outcome finetuning and evaluation.

Annotate with: token prefixes, index date concept, outcome windows, σ_k position in the loss.

Source: Draw in Excalidraw / Inkscape. Export as PDF vector. Use existing architecture sketch from previous sessions as base.

Figure 2 — Survival pair weighting illustration

A 2×2 patient grid illustrating how pair weights are computed:

- Axes: Kaplan-Meier event-mass similarity (x) × censoring support (y).
- Observed events are placed on the KM cumulative primary-event mass scale, so dense event regions get finer resolution than sparse tails.
- Administrative censoring contributes a conditional tail distribution over plausible future event-mass positions rather than a fixed constant.
- Competing deaths are explicit event type 2 and are conservative by default; primary-vs-competing pairs enter only in gamma sensitivity.
- Overlay: DAPT-prior modulation shifts all weights by cohort similarity.

Figure 3 — Embedding space shift (UMAP)

Two-panel UMAP of patient embeddings, coloured by 1-year mortality:

- **Left**: DAPT-stage embeddings (before contrastive training).
- **Right**: OPERA-stage embeddings (after contrastive training).

Show all cohorts in the same plot (use shape for disease, colour for outcome). The expected finding: OPERA separates outcome-similar patients across disease boundaries that DAPT does not.

Script: `opera/visualization/embedding_plots.py` — `plot_embedding_projection(embeddings, labels, method="umap")`.
Run on held-out test set using DAPT and OPERA checkpoints.

Figure 4 — Sigma heatmap

Heatmap: rows = outcomes, columns = cohorts. Cell value = σ_k learned by the multi-cohort OPERA model.

The expected pattern: mortality has low (consistent) σ across cohorts; treatment failure has high variance in σ across cohorts (disease-specific signal). This is the "what the model learned" figure.

Figure 5 — Joint vs. per-cohort performance delta

A dot plot (Cleveland-style) showing ΔAUROC (joint − per-cohort) for each cohort × outcome cell.

- Foundation model variants: expected mostly positive deltas.
- Tabular variants: expected deltas centred near zero or negative.
- Points above the zero line confirm the collaborator's hypothesis.

> The interaction here is the paper's clearest demonstration of why foundation models are structurally better suited to multi-disease registries. Use a side-by-side panel: FM delta (left) vs. tabular delta (right).

Figure 6 – Patient benefit analysis: for whom does OPERA work?

Two-panel figure generated by `opera.run.patient_benefit_analysis`:

- **Panel A**: baseline embedding projection with patient-level Brier gain heatmap for the richer model.
- **Panel B**: baseline-space atypicality versus Brier gain, with LOESS trend and Spearman correlation.

Primary contrasts: general pretraining vs. haematology-only pretraining uses nearest-centroid atypicality; joint OPERA vs. per-cohort OPERA uses within-disease atypicality. This is the mechanistic "for whom" figure: OPERA should help patients who were peripheral in the baseline model's embedding space.

Figure 7 — Clinician uncertainty validation

Scatter plot: model uncertainty (x-axis, embedding variance from MC Dropout) vs. clinician uncertainty rating (y-axis, proportion of reviewers requesting additional information).

Annotate with: Spearman ρ, p-value, n cases.

Expected shape: positive correlation. Highlight 3–5 representative cases in the margin with brief clinical descriptions.

Figure 8 — Label efficiency curves (supplementary)

Learning curves showing AUROC vs. number of labelled training examples for (a) OPERA encoder and (b) tabular XGBoost, evaluated on DLBCL mortality.

The expected finding: OPERA reaches tabular performance with ∼10–20% of the labelled data. This quantifies the label efficiency advantage of pretraining and is a strong practical argument for the approach.

> **How to generate:** subsample training set to [5%, 10%, 20%, 40%, 60%, 80%, 100%] of labels. Run `finetune.py` for each fraction. Plot AUROC vs. fraction, with 95% CI from 3–5 seeds.

Operational Ownership and Registry Coverage

Collaborator-side prerequisites

The collaborator-facing dependency is intentionally small. The remaining items needed from collaborators are:

1. Final laboratory value representation, including cases where a numeric value and a qualification flag must be encoded together.
2. Full-cohort foundation-model pretraining, producing the base checkpoint used by the OPERA ladder.

OPERA-side responsibilities

The OPERA analysis controls cohort directory names, cohort configuration, registry start dates, final outcome parquets, RKKP feature discovery, tabular baselines, supervised finetuning, evaluation, calibration, IPCW metrics, patient-benefit plots, and embedding/prediction visualisations.

Registry coverage rule

Patients whose prediction date precedes reliable registry coverage should remain eligible for MLM/DAPT pretraining if they have pre-index EHR data. They should not, however, be counted as supervised cases, supervised controls, or ordinary RKKP-missing rows for an outcome whose registry was not yet live. In code this is represented by a per-cohort `registry_start_date`; supervised outcome rows with `index_date < registry_start_date` are filtered before binarisation. This keeps the pretraining denominator broad while keeping the supervised denominator scientifically interpretable.

For multi-outcome OPERA, this rule is outcome-specific: a patient may still contribute to other outcome heads if those labels are registry-covered at the prediction date. Coverage denominators must be reported per cohort–outcome cell: number of MLM-eligible patients, number supervised eligible, number registry-ineligible, and the registry coverage fraction.

Analysis Checklist

Step	Description	Status
DAPT embedding store	Run `build_dapt_embedding_store()` once after DAPT checkpoint	
Haematology-only pretraining	Run `opera.run.hematology_pretrain` and include it as a main comparison	
Outcome parquets	Add `time_days` and `event` columns to all outcome parquets	
Registry coverage dates	Fill `registry_start_date` per cohort before supervised finetuning/evaluation	
IPI scores	Compute IPI per cohort, add to `population_full.csv`	
Per-cohort contrastive	Run `contrastive.py` per cohort (DLBCL first)	
Multi-cohort contrastive	Run `contrastive_multicohort.py` on all cohorts jointly	
Per-cohort finetune sweep	Run `sweep.py` with per-cohort OPERA variant	
Joint finetune	Run `joint_finetune.py` — single outcome first, then multi-output	
Joint finetune sweep	Add `opera_joint` variant to sweep, rerun	
Tabular baselines	Train XGBoost per cohort × outcome, export `results/tabular/*.json`	
Sigma analysis	Extract and compare σ_k from contrastive and joint finetune	
Patient benefit analysis	Run `opera.run.patient_benefit_analysis` for the contrast ladder	
Uncertainty extraction	Run `extract_uncertainty()` on held-out set, $N = 30$ passes	
Clinician validation	Present top-100 uncertain cases to haematologists	
UMAP plots	Generate Fig. 3 from DAPT and OPERA checkpoints	
Label efficiency	Run subsampled finetunes for Fig. 8	

Expected Results and Narrative

The paper's argument structure, in order of evidence:

1. **FM beats IPI** across all cohorts and primary outcomes. Establishes clinical relevance. (Table 1)

2. **OPERA > DAPT > Pretrain only** within each cohort. Establishes that each adaptation stage adds something. The DAPT → OPERA gap is our main methodological contribution. (Table 1, all cohorts)

3. **General pretraining > haematology-only pretraining** especially for baseline-atypical haematology patients. Establishes that the "irrelevant" general-population patients contribute reusable clinical structure, not merely scale. (Patient benefit analysis, Figure 6)

4. **FM EHR-only approaches tabular RKKP**. The gap should be small or close on several

outcomes. This is the strong version of the claim: the FM reconstructs RKKP-equivalent structure from longitudinal EHR alone. (Replication of HemaSphere result in FM setting)

5. **Joint training helps FMs, not tabular models**. The ΔAUROC (joint − per-cohort) is positive for FM variants and near-zero or negative for XGBoost. This is the collaborator's hypothesis and the paper's novel empirical finding. (Figure 5, Table 3)

6. **OPERA helps the patients baseline models handle worst**. Patient-level gain should increase with atypicality measured in the baseline embedding space. This provides a mechanistic account of transfer rather than only a model-level performance delta. (Figure 6)

7. σ_k **values are clinically coherent**. Mortality and AKI get low σ (clean, shared signal); treatment failure gets higher σ (disease-specific). Concordance across contrastive and joint finetune stages validates the representation. (Table 4, Figure 4)

8. **Uncertainty tracks clinical difficulty**. Positive correlation between model uncertainty and clinician uncertainty ratings validates that the model has learned what makes a case hard. (Figure 6)

If results are strong enough for Nature Machine Intelligence

The framing shifts from "we adapted a foundation model" to:

"Foundation models pretrained on general EHR data are uniquely suited to joint multi-disease clinical registries because their shared token space naturally encodes cross-disease similarity structure that tabular models require explicit feature engineering to approximate — and outcome-guided adaptation makes this structure measurable and scientifically interpretable."

This requires: (a) the joint FM advantage over tabular is consistent and substantial; (b) the σ_k analysis is interpretable; (c) the clinician uncertainty result holds. Any two of three is sufficient.

Anticipated Reviewer Challenges

Challenge	Response
"Why not train from scratch?"	Label efficiency curves (Fig. 8) show FM reaches tabular performance with $\sim$10–20% of labelled data.
"DALY-CARE is one registry — does it generalise?"	Cross-cohort transfer within registry tests internal generalisation. Cross-disease sigma concordance validates biological coherence.
"Contrastive learning is not novel"	Survival-informed soft pair weighting with censoring-aware reliability is novel. DAPT-prior cross-disease weighting is novel. Combination in a clinical registry is novel.
"The tabular ceiling includes RKKP which FM doesn't have"	This is a feature, not a bug — it quantifies exactly what the FM needs to close. The hybrid model tests whether giving the FM RKKP closes the gap.
"How do you handle censoring?"	Pair weights are computed on Kaplan-Meier cumulative primary-event mass. Administrative censoring contributes a conditional future-event tail distribution; competing deaths are explicit event type 2 and are conservative by default, with gamma sensitivity in the appendix. Report censoring and competing-event fractions per cohort $\times$ outcome.